

Control Centre

Six panels that set up, run and score one test of a trading model on past prices, read left to right. Each saves its own settings, the Scoreboard runs the test with all of them, and every result is kept with the settings that made it.

Does any rule make money after trading costs on a year of prices the model never saw?

A1 Data

which prices, and what counts as a win

Chooses the market, the bar size and the outcome the model is trained to predict.

On the board

Default

Market

crypto

Binance coins, or US stocks from Alpaca

Timeframe

4 hours, test sample

137 coins from LINK on, no bitcoin; for trying settings, not judging a coin

Outcome

barrier

a win when the take-profit comes before the stop; three-way scores up, down or flat instead

Take-profit, Stop

+2, -1 ATR

fixed when the dataset was built; these fields only check it

Horizon

12 bars

two days on 4-hour bars

Crypto, hour

-0.65

Crypto, 4 hour

-0.38

Crypto, day

-1.52

Stocks, hour

-0.01

Stocks, day

+1.15

Median after-cost return a trade on the model's top fifth, overlapping trades removed, 200 random setups, 24 September. Costs were 0.20% for crypto and 0.10% for stocks.
random-suite-20260924-092016.md

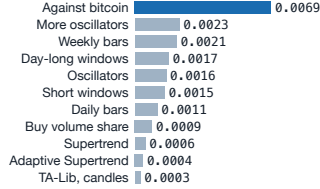
Too few wins

On 4-hour bars the take-profit came first on 25 to 26 per cent of bars, short of the 33.3 per cent that +2 against -1 needs to break even.
univariate-20260921-021638.md

A2 Indicators

the columns the model may use

Chooses which families of computed columns the model is given. None ticked offers them all.



Mean rise in the Brier score when a family's columns were shuffled. 4-hour bars, 40,000 training rows, 8 September.
feature-report-20260908.md

Bitcoin leads

Strength against bitcoin carried three times the weight of the next family, and it is the only family that reads another asset's price.

Useless columns

16 of 90 columns changed the error by less than 0.00001 when shuffled, 9 of them candle patterns.

INDICATOR ENGINES

MACD, moving averages, Fibonacci levels and the confluence score only draw this panel's charts. The model never reads their settings, and none has been tested on prices it had not seen.

B1 Variables

which columns are kept

Can drop weak columns before the fit, with a penalised regression on the training years only. It is off by default.

On the board

Default

Screen first

off

on runs the selection before the fit

L1 mix

1.0

1 is the lasso, which drops columns, 0 is ridge, which only shrinks them, and between is the elastic net

Penalty rule

1se

the strongest penalty within one standard error of the best

Rows

25,000

training rows the selection is fitted on

61 columns offered

46 kept at the best penalty

0 kept at one standard error

Lasso on 25,000 rows of the 1-hour dataset, 21 September.
variable-selection-20260921-0218.md

Not yet measured

Nothing survived the default rule, and no record compares the test-year score with and without the selection.

Single columns

Fitted alone on 4-hour bars, 16 of 33 columns cleared $p < 0.05$, against about 2 expected by chance.
univariate-20260921-021638.md

B2 Training

how the past is split

Holds back the last year for one final test, and cuts the years before it into folds.

On the board

Default

Blind days

365

the test year, scored once at the end

Regime

expanding

the resampling method, walk-forward on a growing window

Folds

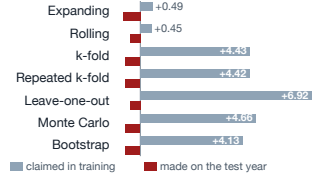
3

cuts of the training years

Purge bars

0

0 to 48 tested; error moved by 0.0001



After-cost return a trade, in per cent, each method claimed for the model it chose, against that model on the test year. Crypto, 4-hour bars.
blind-length-study-20260924-053206.md

Shuffled splits overstate

Methods that ignore time order claimed 4.1 to 6.9 per cent a trade, and the models they chose lost 0.41 to 0.59 per cent on the test year.

Short tests mislead

Split into twelve 30-day stretches, the same test year named 3 to 7 different models as best.

C1 Scoreboard

fit, score and compare

Chooses the models, runs the test, and ranks the result against every fit on record. A fit passes with Theil's U2 below 1 on the test year and an RMSE ratio at or under 1.1; a three-way run is scored on money after cost.

On the board

Default

Models

none

fits three models, logistic regression, random forest and HistGBM, not all six

Class weight

balanced

lifted test U2 to 1.14 to 1.16 on every model that takes it; no weighting did better

Model

Test U2

RMSE ratio

Logistic, elastic net

0.995

1.010

Logistic regression

0.998

1.021

Random forest

1.002

1.073

Classic boosting

1.006

1.180

HistGBM

1.092

4.379

LightGBM

1.093

5.487

Tuning overfitted

Tuning HistGBM cut training RMSE from 0.354 to 0.074 while held-out RMSE rose from 0.484 to 0.517.
model-tuning-histgbm-20260908.md

The record

574 fits, 119 with test U2 below 1, the best 0.980. Four repeats of one setup spread by 0.0062.

C2 Ledger

the paper account, in dollars

Draws the figures a run asks for and reports the paper stock account.

12-1 momentum passed

Hold the 50 US stocks with the best return over the past year, leaving out the latest month. Over 116 months the top tenth beat the market by 1.01 per cent a month ($t = 2.41$).
monthly-factors-20260905.md



The paper account against the same \$100,000 in SPY, the S&P 500 fund, at daily closes.

Marked 23 September, 12:17 New York	
Account, from \$100,000	\$102, 206
Change since 18 Aug	+2.2%
SPY, same period	+0.1%
50 open positions	+\$4, 481
58 closed trades	-\$2, 265
Worst holding, CRS	-20.4%
Cost a fill, 5 to 10 bp assumed	13.2 bp

Paper money

No real money moves, and the live switch is off. Friends' runs on the public page left 7 paper tickets on 25 September, none settled yet.

Testing an idea	1 Change one thing Change one setting on its panel and save it. Two changes at once cannot be told apart.	2 Run the test Press Run the test on C1, which uses every saved setting.	3 Read the marks Test U2 below 1, RMSE ratio at or under 1.1, and for the three-way outcome, money after cost.	4 Compare Against the run before. A change in U2 smaller than 0.0062 is within the spread of repeats.	5 Keep or revert Keep what beat the record. Never choose a setting by looking at the test year, because it has then been seen.		
Terms	ATR Average true range, the typical high-to-low move over 14 bars. RMSE Root mean squared error. Held-out over training RMSE is the RMSE ratio, or overfit ratio.	Triple barrier A trade ends at the take-profit, the stop or the time limit, whichever comes first. Top fifth The fifth of bars the model rated most likely to rise.	Base rate The share of bars where the take-profit came first. Brier score The mean squared error of a stated probability.	Test year The final 365 days, held back and scored once, the board's blind period. MACD A fast moving average of price minus a slow one.	Fold One cut of the training years, fitted on part and scored on the rest. Gradient boosting Trees fitted in turn, each to the last one's errors. HistGBM and LightGBM are two kinds.	Walk-forward Fit on the past, score the stretch after it, then move forward in time. Basis point One hundredth of a per cent, written bp.	Theil's U2 The model's error over that of always forecasting the base rate. Below 1 beats it. Paper account Real prices and simulated orders, with no money at risk.

ANSWER SO FAR

Crypto, no. Of 120 random crypto test setups run on 24 September, 9 made money after costs once overlapping trades were removed, and hourly, 4-hour and daily bars all lost at the median.

US stocks, mostly the market. Daily stock setups made money in 43 of 45, but buying every stock made money in all 45, a median 0.71 per cent a trade, and the model's picks added a median of only 0.18.

One strategy passed. 12-1 momentum on US stocks cleared every test and has run on a paper account since 18 August.